

AUTOMATIK

Database Impact Summary — REVISION Modules

Generated: June 10, 2026

Database: automatik (26 existing tables)

Modules: M&R · Procurement · F&I

10

New Tables

6

Altered Tables

20

Unchanged Tables

3

Modules

1. MAINTENANCE & REPAIR (M&R)

1.1 Tables to ADD

■ repair_orders

Active Repair Order workspace — core entity tracking vehicle through service lifecycle.

Column	Type	Constraints / Notes
ro_id	INT	PK, AUTO_INCREMENT
ro_number	VARCHAR(20)	UNIQUE — e.g. RO-1001
booking_id	INT	FK → service_bookings ON DELETE SET NULL
vehicle_id	INT	NOT NULL, FK → vehicles
customer_id	INT	FK → customer_details ON DELETE SET NULL
service_advisor_id	INT	FK → users (SA who opened RO)
assigned_technician_id	INT	FK → users (agent/technician role)
customer_narrative	TEXT	Read-only copy of issue description
status	ENUM	draft awaiting_signature in_progress pending_supplement ready_for_testing ready_for_checkout completed cancelled
progress_pct	DECIMAL(5,2)	Calculated: completed items / total items
total_parts_cost	DECIMAL(12,2)	Aggregated from line items
total_labor_cost	DECIMAL(12,2)	Aggregated from line items
total_misc_cost	DECIMAL(12,2)	Aggregated from line items
sub_total	DECIMAL(12,2)	parts + labor + misc
sales_tax	DECIMAL(12,2)	Calculated tax liability
grand_total	DECIMAL(12,2)	sub_total + sales_tax
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP

Spec ref: §2.2 Active RO Workspace · §3.1 Intake & Estimate · §4.1 Invoice Visualization

■ ro_line_items

Parts, Labor, and Misc line items — serves as both estimate grid and final invoice breakdown.

Column	Type	Constraints / Notes
line_item_id	INT	PK, AUTO_INCREMENT
ro_id	INT	NOT NULL, FK → repair_orders ON DELETE CASCADE
line_type	ENUM	part labor miscellaneous
description	VARCHAR(255)	
sku	VARCHAR(50)	Nullable; typeahead-linked to supplies.part_number
supply_id	INT	FK → supplies ON DELETE SET NULL
qty	DECIMAL(10,2)	Float for labor hours, INT for parts
unit_cost	DECIMAL(10,2)	Auto-populated from supplies or manual
total_cost	DECIMAL(12,2)	qty * unit_cost
is_authorized	TINYINT(1)	DEFAULT 0; toggled on customer sign-off
sort_order	INT	Display ordering
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Spec ref: §3.1 Financial Estimator Grid · §4.1 Invoice Visualization

■ ro_supplemental_issues

Technician-reported supplemental issues. Suspends RO and notifies Service Advisor on submit.

Column	Type	Constraints / Notes
issue_id	INT	PK, AUTO_INCREMENT
ro_id	INT	NOT NULL, FK → repair_orders ON DELETE CASCADE
title	VARCHAR(200)	
description	TEXT	
estimated_cost	DECIMAL(12,2)	
status	ENUM	pending approved rejected — DEFAULT 'pending'
resolved_by	INT	FK → users ON DELETE SET NULL
resolution_notes	TEXT	
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
resolved_at	TIMESTAMP	NULL

Spec ref: §3.2 Supplemental Issue Protocol

1.2 Tables to ALTER

■ service_bookings

Change	Detail
Add pickup_type	ENUM('dealer_pickup','customer_dropoff') DEFAULT NULL
Add service_classification	ENUM('maintenance','repair') DEFAULT NULL — badge indicator §2.1
Add assigned_technician_id	INT, FK → users ON DELETE SET NULL
Alter status ENUM	Add: in_progress awaiting_signature pending_supplement ready_for_testing ready_for_checkout
Add ro_id	INT, FK → repair_orders ON DELETE SET NULL

■ payments

Change	Detail
Alter payment_method ENUM	Add 'digital_wallet' — spec §4.2 Cash Credit Digital Wallet

■ notifications

No schema change needed — ref_type already VARCHAR(50).

New ref_type values: 'repair_order', 'supplemental_issue'

1.3 Tables UNCHANGED by M&R

access_tokens · agent_commissions · agent_details · agent_tasks · amortization_schedule · audit_logs · chatbot_logs · customer_details · documents · inquiries · insurance_records · loan_details · sales · sales_contracts · service_slots · suppliers · supplies · system_settings · user_profile · users · vehicle_photos · vehicles · warranty_claims

2. PROCUREMENT

2.1 Tables to ADD

■ purchase_orders

Central PO workspace — acquisition lifecycle from draft to fulfilment.

Column	Type	Constraints / Notes
po_id	INT	PK, AUTO_INCREMENT
po_number	VARCHAR(20)	UNIQUE — e.g. PO-5001
supplier_id	INT	NOT NULL, FK → suppliers (accredited only)
order_value	DECIMAL(12,2)	Financial commitment total
status	ENUM	draft sent_to_supplier partially_fulfilled completed cancelled
dispatched_at	TIMESTAMP	NULL; set on dispatch
created_by	INT	FK → users
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP

Spec ref: §2.2 PO Workspace · §3.1 Purchase Order Builder

■ purchase_order_items

Line-item grid — references vehicles (dealer trades) or supplies/parts.

Column	Type	Constraints / Notes
item_id	INT	PK, AUTO_INCREMENT
po_id	INT	NOT NULL, FK → purchase_orders ON DELETE CASCADE
line_type	ENUM	vehicle supply
vehicle_id	INT	FK → vehicles ON DELETE SET NULL
supply_id	INT	FK → supplies ON DELETE SET NULL
description	VARCHAR(255)	Free-text or auto-populated
make	VARCHAR(50)	Vehicle line items
model	VARCHAR(100)	Vehicle line items
year	SMALLINT	Vehicle line items
color	VARCHAR(50)	Vehicle line items
qty_ordered	INT	
unit_cost	DECIMAL(10,2)	
total_cost	DECIMAL(12,2)	qty_ordered * unit_cost

Spec ref: §3.1 Line-Item Grid (Vehicles/Supplies)

■ receiving_records

Receiving & Inspection Bay — tracks physical deliveries against POs.

Column	Type	Constraints / Notes
delivery_id	INT	PK, AUTO_INCREMENT
delivery_number	VARCHAR(20)	UNIQUE — e.g. DEL-3001
po_id	INT	NOT NULL, FK → purchase_orders
inspection_status	ENUM	pending inspected_approved inspected_discrepancy
received_by	INT	FK → users
received_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
notes	TEXT	
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Spec ref: §2.3 Receiving & Inspection Bay · §3.2 Inspection Checklist

■ inspection_items

Physical count and quality condition per PO line item during receiving.

Column	Type	Constraints / Notes
inspection_item_id	INT	PK, AUTO_INCREMENT
delivery_id	INT	NOT NULL, FK → receiving_records ON DELETE CASCADE
po_item_id	INT	NOT NULL, FK → purchase_order_items
expected_qty	INT	From PO
received_qty	INT	Physical count entered by clerk
condition	ENUM	excellent damaged_defective — mandatory
notes	TEXT	
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Spec ref: §3.2 Physical Count Input · Quality Flagging

■ supplier_invoices

Financial Settlement Ledger — invoice matching and payment clearance.

Column	Type	Constraints / Notes
invoice_id	INT	PK, AUTO_INCREMENT
po_id	INT	NOT NULL, FK → purchase_orders
supplier_id	INT	NOT NULL, FK → suppliers
invoice_number	VARCHAR(100)	Supplier-provided reference
invoice_amount	DECIMAL(12,2)	Total on supplier invoice
po_total	DECIMAL(12,2)	Original PO total (three-way match display)
inspection_total	DECIMAL(12,2)	Receiving total (three-way match display)
payable_amount	DECIMAL(12,2)	Amount after matching
match_status	ENUM	match_ok discrepancy
payment_status	ENUM	unpaid processing cleared — DEFAULT 'unpaid'
paid_at	TIMESTAMP	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Spec ref: §2.4 Financial Settlement Ledger · §4.1 Invoice Matching

2.2 Tables to ALTER

■ suppliers

Change	Detail
Add catalog_url	VARCHAR(500) NULL — supplier's digital catalog link
Add lto_expiry	DATE NULL — LTO/agreement expiry; drives Green/Red accreditation indicator §2.1

■ vehicles

Change	Detail
Add 'incoming' status (optional)	New ENUM value — represents vehicles ordered via PO but not yet received. Current: available reserved discontinued delivered

■ supplies

No schema change required. stock_qty already exists. PO → Receiving → Inspection workflow updates stock_qty atomically on inspection approval (§4.2).

2.3 Tables UNCHANGED by Procurement

access_tokens · agent_commissions · agent_details · agent_tasks · amortization_schedule · audit_logs · chatbot_logs · customer_details · documents · inquiries · insurance_records · loan_details · notifications · payments · sales · sales_contracts · service_bookings · service_slots · system_settings · user_profile · users · vehicle_photos · warranty_claims

3. FINANCING & INSURANCE (F&I)

3.1 Tables to ADD

■ loan_applications

Credit application triage pipeline — pre-sale entity from submission through underwriting decision.

Column	Type	Constraints / Notes
application_id	INT	PK, AUTO_INCREMENT
application_number	VARCHAR(20)	UNIQUE — e.g. LA-4001
customer_id	INT	NOT NULL, FK → users
vehicle_id	INT	FK → vehicles (collateral)
status	ENUM	pending_review approved rejected
full_name	VARCHAR(150)	Applicant demographics
national_id	VARCHAR(50)	Tax ID / National ID
employment_status	VARCHAR(50)	Employed Self-Employed OFW etc.
gross_monthly_income	DECIMAL(12, 2)	Verified income
vehicle_msrp	DECIMAL(12, 2)	Manufacturer's Suggested Retail Price
down_payment	DECIMAL(12, 2)	Intended downpayment
requested_principal	DECIMAL(12, 2)	Auto-calc: MSRP - down_payment
proposed_amortization	DECIMAL(12, 2)	Proposed monthly payment
dti_ratio	DECIMAL(5, 2)	$\text{proposed_amortization} / \text{gross_monthly_income} * 100$
risk_tier	ENUM	tier_a tier_b tier_c
reviewed_by	INT	FK → users ON DELETE SET NULL
rejection_reason	VARCHAR(255)	Required if status = 'rejected'
rejection_notes	TEXT	Optional narrative
reviewed_at	TIMESTAMP	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Spec ref: §2.2 Loan Applications Queue · §3.1 Split-Panel Credit Assessment Interface

■ financing_contracts

Registry between credit approval and amortization generation. Tracks finalized terms pre-first-payment.

Column	Type	Constraints / Notes
contract_id	INT	PK, AUTO_INCREMENT
contract_number	VARCHAR(20)	UNIQUE — e.g. FC-2001
loan_application_id	INT	FK → loan_applications ON DELETE SET NULL
sale_id	INT	FK → sales ON DELETE SET NULL
customer_id	INT	NOT NULL, FK → users
principal_balance	DECIMAL(12,2)	vehicle_base_price - confirmed_down_payment
annual_interest_rate	DECIMAL(5,2)	0.00 – 30.00
term_months	TINYINT	12 24 36 48 60
monthly_amortization	DECIMAL(12,2)	Standard amortization formula
insurance_policy_id	INT	FK → insurance_records ON DELETE SET NULL
monthly_insurance_addon	DECIMAL(10,2)	If roll-premium toggle TRUE
total_monthly_commitment	DECIMAL(12,2)	monthly_amortization + monthly_insurance_addon
status	ENUM	draft active paid_off defaulted
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
signed_at	TIMESTAMP	NULL

Spec ref: §2.2 Financing Contracts Workspace · §3.2 Financing Configuration

3.2 Tables to ALTER

■ insurance_records

Change	Detail
Add premium_cost	DECIMAL(10,2) NOT NULL DEFAULT 0.00 — annualized policy cost
Add roll_into_amortization	TINYINT(1) NOT NULL DEFAULT 0 — roll premium into monthly amortization
Add provider_entity	VARCHAR(150) — partner insurance carrier name
Alter status ENUM	Add 'unverified' and 'bound'. Extended: unverified active bound expired cancelled

■ loan_details

Change	Detail
Add application_id	INT, FK → loan_applications ON DELETE SET NULL
Add contract_id	INT, FK → financing_contracts ON DELETE SET NULL

■ user_profile (optional — or keep in loan_applications only)

Change	Detail
Add national_id	VARCHAR(50) NULL — tax/national ID
Add employment_status	VARCHAR(50) NULL
Add gross_monthly_income	DECIMAL(12,2) NULL

■ notifications

No schema change. ref_type accepts new values:

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'loan_application' · 'financing_contract' · 'insurance_policy'
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3.3 Tables UNCHANGED by F&I

access_tokens · agent_commissions · agent_details · agent_tasks · amortization_schedule · audit_logs · chatbot_logs ·
customer_details · documents · inquiries · payments · sales · sales_contracts · service_bookings · service_slots · suppliers · supplies ·
system_settings · vehicle_photos · vehicles · warranty_claims

4. ROLLUP — ALL MODULES COMBINED

4.1 New Tables (10 total)

#	Table	Module	Purpose
1	repair_orders	M&R;	Active Repair Order lifecycle
2	ro_line_items	M&R;	Estimate & invoice line items
3	ro_supplemental_issues	M&R;	Technician-reported supplementals
4	purchase_orders	Procurement	PO drafting & fulfilment
5	purchase_order_items	Procurement	PO line-item grid
6	receiving_records	Procurement	Receiving & inspection tracking
7	inspection_items	Procurement	Physical count & condition per item
8	supplier_invoices	Procurement	Invoice matching & payment
9	loan_applications	F&I;	Credit application triage pipeline
10	financing_contracts	F&I;	Post-approval contract registry

4.2 Altered Tables (6 existing)

#	Table	Module	Changes
1	service_bookings	M&R;	Add pickup_type, service_classification, assigned_technician_id; extend status ENUM; add ro_id
2	payments	M&R;	Extend payment_method ENUM → add digital_wallet
3	suppliers	Procurement	Add catalog_url, lto_expiry
4	insurance_records	F&I;	Add premium_cost, roll_into_amortization, provider_entity; extend status ENUM
5	loan_details	F&I;	Add FK → loan_applications; FK → financing_contracts
6	user_profile	F&I;	Add national_id, employment_status, gross_monthly_income (optional)

4.3 Unchanged Tables (20)

access_tokens · agent_commissions · agent_details · agent_tasks · amortization_schedule · audit_logs · chatbot_logs · customer_details · documents · inquiries · notifications (ref_type values only) · sales · sales_contracts · service_slots · supplies · system_settings · users · vehicle_photos · vehicles · warranty_claims

5. ENTITY-RELATIONSHIP NOTES

5.1 Cross-Module Dependency Diagram

loan_applications ■■→ financing_contracts ■■→ sales ■■→ sales_contracts ↓ insurance_records ↓
amortization_schedule ■■→ payments service_bookings ■■→ repair_orders ■■→ ro_line_items ↓
ro_supplemental_issues purchase_orders ■■→ purchase_order_items ■■→ supplies / vehicles ↓
receiving_records ■■→ inspection_items ↓ supplier_invoices ■■→ suppliers

5.2 Key Dependency Rules

Rule	Description
1	An RO must have a service_bookings record (booking comes first, then intake creates the RO).
2	A PO can reference either vehicles OR supplies via its line items, not both per item.
3	A receiving_record is always linked to a PO; a PO can have multiple partial deliveries.
4	supplier_invoices depends on three-way match: PO total ↔ Inspection total ↔ Invoice total.
5	loan_applications precedes financing_contracts, which may optionally precede sale creation.
6	insurance_records is linked from financing_contracts via insurance_policy_id.